

COMPACT-RANGE VIETORIS POWERS OF COUNTABLE ORDINALS

LUCAS ALVES DE ALMEIDA

lucasdeirva8x@gmail.com

ABSTRACT. For every ordinal $\omega < \alpha < \omega_1$, the ordered compact-set space $K(\alpha, \text{ord})$ of Caruvana and Holshouser is second-countable and Lindelöf, but not σ -compact, resolving Question 1 of arXiv:2507.17936v3. The central observation is that closed inclusions induce closed embeddings of compact-range Vietoris powers when the index is fixed. Since $[0, \omega]$ is closed in every such α , the author's earlier non- σ -compactness theorem for $\omega + 1$ immediately transfers to the entire interval. Lindelöfness follows from a general countable-base theorem. We also retain a self-contained delayed diagonal proof, which strengthens the negative conclusion to failure of the Menger property. The principal ordinal theorem has a Lean 4 formalization and a reproducible local verification record.

Key words and phrases. Vietoris power; ordered compact sets; countable ordinal; Lindelöf space; σ -compactness; Menger property; formal verification.

1. EARLIER WORK AND THE PRESENT RESULT

Caruvana and Holshouser [1, Question 1, p. 11] ask whether $K(\alpha, \text{ord})$ is σ -compact or Lindelöf for $\omega < \alpha < \omega_1$, explicitly singling out $\alpha = \omega + 1$. Throughout this paper, page and question numbers refer to arXiv version 3.

In the earlier manuscript *On the Vietoris Power of a Convergent Sequence* [2], Lucas Alves de Almeida established that $K(\omega + 1, \text{ord})$ is second-countable and Lindelöf but not σ -compact, with an accompanying Lean verification. Its negative argument uses finite occurrence bounds and a delayed diagonal. Once this base case is known, the extension of non- σ -compactness follows directly from a closed embedding; no new diagonal argument is needed for that transfer.

Theorem 1.1. For every ordinal $\omega < \alpha < \omega_1$, the space $K(\alpha, \text{ord})$ is second-countable, hence Lindelöf, and is neither σ -compact nor Menger.

The proof has two structural ingredients: a countable-base theorem and preservation of closed inclusions by the compact-range construction at a fixed index. Section 3 applies the second ingredient to $[0, \omega] \subseteq \alpha$ and [2]. Section 4 supplies an independent diagonal proof and the non-Menger strengthening. Both successor and limit ordinals are covered; compactness of α itself is unnecessary.

2. THE SPACE AND ITS COUNTABLE BASE

For a space X and a cardinal κ , the Vietoris-power topology has basic sets [1, Definition 1.8 and Proposition 1.9, p. 4]

$$[U; F, V] = \{f \in X^\kappa : \text{img}(f) \subseteq U, f(i) \in V_i \ (i \in F)\},$$

where U and all V_i are open and $F \subseteq \kappa$ is finite. Definition 1.14 [1, p. 5] gives the compact-range subspace

$$K(X, \text{ord}, \kappa) = \{f : \kappa \rightarrow X : \text{img}(f) \text{ is compact}\}.$$

The default index for $K(X, \text{ord})$ is the supremum of the cardinalities of the nonempty compact subsets of X , plus ω in cardinal arithmetic. For a countable ordinal α , this index is ω . Accordingly, write

$$K_\alpha = \{f : \omega \rightarrow \alpha : \text{img}(f) \text{ is compact}\}.$$

The functions need not be continuous or order-preserving. Compactness here refers to the actual image, rather than only to its closure. We use the open tube and coordinate sets

$$T(U) = \{f \in K_\alpha : \text{img}(f) \subseteq U\},$$

$$E(i, U) = \{f \in K_\alpha : f(i) \in U\}.$$

Thus a basic neighborhood is a tube intersected with finitely many coordinate restrictions. In particular, every coordinate evaluation $e_i(f) = f(i)$ is continuous.

Lemma 2.1. If X is second-countable, then $K(X, \text{ord}, \omega)$ is second-countable.

Proof. Let B be a countable base for X , and let $B^\cup$ be the countable family of finite unions of its members. Consider the basic sets with range bound $W \in B^\cup$, finite coordinate set $F \subseteq \omega$, and coordinate bounds $B_i \in B$. There are countably many such choices.

Given $f \in [U; F, V]$, cover $\text{img}(f)$ by members of B contained in U . Compactness of $\text{img}(f)$ gives finitely many such members with union W , so that $\text{img}(f) \subseteq W \subseteq U$. For each $i \in F$, choose $B_i \in B$ with $f(i) \in B_i \subseteq V_i$. Then

$$f \in [W; F, B] \subseteq [U; F, V].$$

The indicated countable family is therefore a base. $\square$

Corollary 2.2. For every countable ordinal α , the space K_α is second-countable and Lindelöf.

Proof. The standard interval and ray base of a countable ordinal has only countably many possible endpoints. Apply Lemma 2.1, and then the implication from second countability to Lindelöfness. $\square$

The compact-image hypothesis in Lemma 2.1 is essential to this proof. No corresponding claim is made here for the full Vietoris power $V(X^\omega)$.

3. CLOSED EMBEDDINGS AND COUNTABLE COMPACT COVERS

Proposition 3.1 (closed-subspace transfer). If A is closed in X , the inclusion induces a homeomorphism of $K(A, \text{ord}, \omega)$ onto the closed subspace $Z_A = \{f \in K(X, \text{ord}, \omega) : \text{img}(f) \subseteq A\}$.

Proof. A subset of A is compact in the subspace topology exactly when it is compact in X . Therefore the coordinatewise inclusion identifies the two carriers in the statement. The set Z_A is closed, since it is the intersection over $k \in \omega$ of the closed conditions $f(k) \in A$. This is also the range-only case of [1, Proposition 1.12].

Under this identification, an ambient basic set with open bounds U and V_i restricts to the basic set with bounds $U \cap A$ and $V_i \cap A$. Conversely, every open bound in the subspace A is the intersection of A with an open subset of X . The two induced topologies consequently agree. $\square$

Corollary 3.2. If A is closed in X and $K(A, \text{ord}, \omega)$ is not σ -compact, then $K(X, \text{ord}, \omega)$ is not σ -compact.

Proof. A closed subspace of a σ -compact space is σ -compact: intersect a countable compact cover with the closed subspace. Apply Proposition 3.1 and take the contrapositive. $\square$

Corollary 3.3 (the full ordinal interval). For every $\omega < \alpha < \omega_1$, the space $K(\alpha, \text{ord})$ is not σ -compact.

Proof. The initial segment $A = [0, \omega]$ is closed in α and homeomorphic to $\omega + 1$. The default indexing cardinals for A and α are both ω , by Section 2. Thus Proposition 3.1 identifies $K(\omega + 1, \text{ord})$ with a closed subspace of $K(\alpha, \text{ord})$. The earlier theorem of Almeida [2] and Corollary 3.2 give the conclusion. $\square$

The equality of the indexing cardinals is part of this application. Proposition 3.1 compares powers with the same index; it does not assert that default-index powers preserve every closed inclusion. Corollary 2.2 and Corollary 3.3 already answer both parts of Question 1 throughout its stated interval.

4. A SELF-CONTAINED DIAGONAL AND THE Menger STRENGTHENING

For completeness, we now recover Corollary 3.3 without invoking [2] as a theorem. This also supplies the explicit open covers needed to prove the stronger non-Menger conclusion.

Fix $\omega < \alpha < \omega_1$ and put $A = [0, \omega] \subseteq \alpha$. The subspace A is homeomorphic to $\omega + 1$, so it is compact. It is closed in α , since its complement is an open final segment. Define

$$S_\alpha = \{f \in K_\alpha : f(0) = \omega, \text{img}(f) = A\}.$$

Lemma 4.1. The set S_α is closed in K_α .

Proof. The condition $f(0) = \omega$ is closed by continuity of evaluation. The condition $\text{img}(f) \subseteq A$ is the intersection, over $k \in \omega$, of the closed conditions $f(k) \in A$. For each $a \in A$, the condition $a \in \text{img}(f)$ is closed because its complement is the open tube $T(\alpha \setminus \{a\})$. Singletons are closed in the Hausdorff ordinal space. Intersecting these conditions proves the assertion. $\square$

Lemma 4.2. If $C \subseteq S_\alpha$ is compact and $n \in \omega$, then there is $b \in \omega$ such that each $f \in C$ takes the value n at some coordinate $k \leq b$.

Proof. Each finite ordinal n is isolated in α , so the sets $E(k, \{n\})$, indexed by $k \in \omega$, are open. They cover C because every member of S_α has image A . A finite subcover exists. Take b to be the largest of the selected indices, or $b = 0$ when C is empty. $\square$

Proposition 4.3 (direct diagonal proof). For every $\omega < \alpha < \omega_1$, the space K_α is not σ -compact.

Proof. Suppose K_α has a countable compact cover. By Lemma 4.1, intersecting its members with S_α gives compact sets C_n with $S_\alpha = \bigcup_{n \in \omega} C_n$. Apply Lemma 4.2 to C_n and the value n , obtaining a bound $b(n)$.

Choose distinct positive integers $j(n) > b(n)$. One explicit choice is

$$j(0) = b(0) + 1, j(n + 1) = \max\{j(n), b(n + 1)\} + 1.$$

Define $g : \omega \rightarrow \alpha$ by

$$g(j(n)) = n \ (n \in \omega), g(k) = \omega \ (k \notin \{j(n) : n \in \omega\}).$$

The function has $g(0) = \omega$ and image exactly A , so $g \in S_\alpha$ and its image is compact. For each n , its only occurrence of the value n is at $j(n) > b(n)$; hence $g \notin C_n$. This contradicts the cover. $\square$

The fixed image A is the only structural addition needed to the earlier diagonal proof in [2]. Attempting instead to enumerate the entire ordinal would fail to give an element of K_α when α is noncompact.

Theorem 4.4. For every $\omega < \alpha < \omega_1$, the space K_α is not Menger.

Proof. A space is Menger when, for every sequence of open covers, finite selections from the respective covers can be made whose union covers the space. For $n, m \in \omega$, put

$$H_{n,m} = (K_\alpha \setminus S_\alpha) \cup \bigcup_{k \leq m} E(k, \{n\}).$$

For each n , the increasing family $U_n = \{H_{n,m} : m \in \omega\}$ is an open cover: points outside S_α belong to every member, and points in S_α eventually take the value n . Given finite selections from the U_n , choose $b(n)$ large enough to bound the indices of all selected members of U_n , taking $b(n) = 0$ for an

empty selection. The diagonal function from Proposition 4.3, with $j(n) > b(n)$, belongs to none of the selected sets. Thus these selections cannot cover K_α . $\square$

Corollaries 2.2 and 3.3, together with Theorem 4.4, prove Theorem 1.1.

5. THE ORDINAL CLASSIFICATION

Together with the discrete cases and Proposition 2.17 of [1], the result gives the following classification under the default indexing convention:

Ordinal α	Lindelöf	σ -compact	Menger
$0 \leq \alpha \leq \omega$	Yes	Yes	Yes
$\omega < \alpha < \omega_1$	Yes	No	No
$\alpha \geq \omega_1$	No	No	No

For $\alpha \leq \omega$, the ground space is countable and discrete, and every compact image is finite. Fixing an initial segment that lists every value of the image gives a compact product of a finite discrete space on the remaining coordinates. Countably many such compact subspaces cover K_α . The case $\alpha = 0$ is empty. For uncountable ordinals, Proposition 2.17 of [1] gives failure of Lindelöfness, which also precludes the other two covering properties.

6. COMPATIBILITY WITH THE SOURCE PAPER

Theorem 2.16(d) of [1] treats the endpoint $\alpha = \omega$: compact images in this discrete space are finite. The negative result above starts strictly after that endpoint. Proposition 2.17 of [1] treats uncountable ordinals, outside the interval answered here. Theorem 2.18 concerns the full power $V(\omega^\omega)$, which also includes functions with noncompact image; it therefore does not conflict with our countable-base theorem for the compact-range subspace.

The fixed index in Lemma 2.1 is essential. Example 2.33 of [1] proves that $K(\mathbb{R}, \text{ord})$ is not Lindelöf using the default uncountable index, the cardinality of the continuum. Lemma 2.1 instead concerns $K(\mathbb{R}, \text{ord}, \omega)$, so these conclusions are compatible. Finally, the continuous range map in Proposition 2.26 sends the ordered space to the ordinary Vietoris hyperspace. Covering properties of that image do not in general lift to its domain; this causes no conflict with our non-Menger conclusion.

The main proof uses the defining basic neighborhoods, the closed-subspace argument and [2]; Section 4 removes even the dependence on [2] as an assumed result. No compact-covering-map assertion from Section 2.4 of [1] is needed.

7. FORMAL VERIFICATION AND SOURCE CORRESPONDENCE

The accompanying Lean 4 project uses actual mathlib ordinals. For $a : \text{Ordinal}$, the ground space is the subtype $\{b : \text{Ordinal} : b < a\}$, with the topology inherited from the ordinal order topology. Its convexity identifies this with its own order topology. The compact-range carrier is defined by actual compactness of the range, and its topology is induced from the topology generated by the source basic sets. The formal theorem quantifies over every $\omega < a < \omega_1$.

The formalization proves the general second-countability lemma and realizes the fixed compact image as $[0, \omega]$ inside the ordinal subtype. It establishes closedness of the surjection family and obtains occurrence bounds from finite subcovers. The delayed sequence uses the natural-number pairing bijection, as in [2]; distinct positions $j(n) = \text{pair}(n, b(n) + 1)$ lie beyond the corresponding bounds.

Checked principal declaration: `VietorisOrdinals.ordinal_full`

The standard Menger property is defined using arbitrary open covers and finite selections. The formal proof constructs the covers from Theorem 4.4 and verifies that every finite selection misses a delayed diagonal function. The checked proof follows the self-contained argument in Section 4. Proposition 3.1 and its abstract transfer corollary are proved in this manuscript but are not separately formalized in the supplied Lean project.

The compiler is Lean 4.19.0 with mathlib v4.19.0. The transitive axiom audit for the principal theorem reports only `propext`, `Classical.choice` and `Quot.sound`. The verification record includes exact theorem types, compiler output, dependency revisions and source hashes. It is a local reproducible verification record, not independent peer review.

The numerical reduction of the authors' default indexing cardinal to ω follows from countability, as explained in Section 2. Identifying the notation and question in the source PDF with the formal definitions is an explicit mathematical interpretation; it is not a task performed by the Lean kernel. The accompanying verification record states the precise scope of every checked result.

AI DISCLOSURE

This work was developed collaboratively by Lucas Alves de Almeida and ChatGPT-Astra. Almeida proposed extending the earlier result, directed the investigation toward the delayed diagonal method, and guided the work. ChatGPT-Astra developed the main general argument and performed the primary mathematical development, Lean formalization, verification work, and manuscript preparation under Almeida's direction. This disclosure does not assert independent mathematical review.

REFERENCES

- [1] Christopher Caruana and Jared Holshouser, An Adaptation of the Vietoris Topology for Ordered Compact Sets, arXiv:2507.17936v3 (10 November 2025). Definitions 1.8 and 1.14, pp. 4-5; Theorem 2.16; Proposition 2.17 and Question 1, p. 11. <https://arxiv.org/abs/2507.17936v3>
- [2] Lucas Alves de Almeida, On the Vietoris Power of a Convergent Sequence, revised unpublished manuscript supplied by the author, 2026, 3 pp., with accompanying Lean 4 verification project.